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## What is Matplotlib?
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** Matplotlib is a Python library used for data visualization.
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** Data visualization means representing data using charts and graphs.
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** Matplotlib is mainly used to create:
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- 1) Line charts
- 2) Bar charts
- 3) Pie Charts
- 4) Scatter Plots
- 5) Histograms

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=====
## Why Matplotlib?
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** Matplotlib is used for:
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- 1) Visualizing data
- 2) Understand data easily
- 3) Comparing values
- 4) Finding Trends
- 5) Finding Patterns
- 6) Create Reports
- 7) Data Analysis

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## Matplotlib Installation
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```
=> Matplotlib is an external library.
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```
=> We need to install it using pip.
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```
Command : pip install matplotlib
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```
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import matplotlib.pyplot as plt
```

```
months = ["Jan", "Feb", "Mar", "Apr"]
sales = [10000, 15000, 12000, 18000]
```

```
plt.plot(months, sales, marker="o", linestyle="--", color="blue")
```

```
plt.title("Sales by Month")
```

```
plt.xlabel("Month")
plt.ylabel("Sales")
```

```
plt.show()
```

```
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import matplotlib.pyplot as plt
```

```
courses = ["Python", "Java", "DevOps", "AWS"]
students = [50, 40, 30, 35]
```

```
#plt.bar(courses, students)
plt.barh(courses, students)
```

```
plt.title("Course Enrollment Report")
plt.xlabel("Course")
plt.ylabel("Students Count")
```

```
plt.show()
```

```
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import matplotlib.pyplot as plt
```

```
courses = ["Python", "Java", "DevOps", "AWS"]
students = [50, 40, 30, 35]
```

```
plt.pie(students, labels=courses, autopct="%1.1f%%")
```

```
plt.title("Course Enrollment Share")
```

```
plt.show()
```

```
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import matplotlib.pyplot as plt
```

```
study_hours = [1, 2, 3, 4, 5]
marks = [35, 45, 60, 75, 90]
```

```
plt.scatter(study_hours, marks)
```

```
plt.title("Study Hours vs Marks")
plt.xlabel("Study Hours")
plt.ylabel("Marks")
```

```
plt.show()
```

```
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import matplotlib.pyplot as plt
```

```
marks = [35, 45, 50, 60, 65, 70, 75, 80, 85, 90, 95]
```

```
plt.hist(marks)
```

```
plt.title("Marks Distribution")
plt.xlabel("Marks")
plt.ylabel("Number of Students")
```

```
plt.show()
```

```
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import matplotlib.pyplot as plt  
  
months = ["Jan", "Feb", "Mar", "Apr"]  
  
python_sales = [10000, 15000, 12000, 18000]  
java_sales = [8000, 12000, 10000, 14000]  
  
plt.plot(months, python_sales, marker="o", label="Python")  
plt.plot(months, java_sales, marker="o", label="Java")  
  
plt.title("Course Sales Comparison")  
plt.xlabel("Months")  
plt.ylabel("Sales")  
  
plt.legend()  
  
plt.grid()  
  
plt.show()
```

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Important Points

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- ** Matplotlib is used for data visualization.
- ** Data visualization means representing data using charts.
- ** Matplotlib is an external library.
- ** Install Matplotlib using "pip install matplotlib".
- ** pyplot is commonly used module in Matplotlib.
- ** Import pyplot using "import matplotlib.pyplot as plt".
- ** plot() creates line chart.
- ** bar() creates bar chart.
- ** barh() creates horizontal bar chart.
- ** pie() creates pie chart.
- ** scatter() creates scatter plot.
- ** hist() creates histogram.
- ** show() displays chart.
- ** title() adds chart title.
- ** xlabel() adds x-axis label.
- ** ylabel() adds y-axis label.
- ** legend() displays labels.

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** grid() displays grid lines.
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** Matplotlib is useful for Data Science, Machine Learning and reports.
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